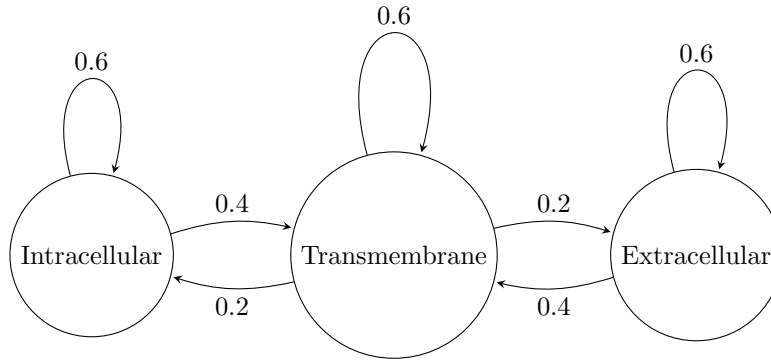


BENG 203 Assignment 3

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Problem 1:



For this Hidden Markov Model, there are three hidden states $Q = \{Intracellular, Transmembrane, Extracellular\}$. The observed states Σ are the 20 amino acids. The initial state distribution is $\pi = \{1, 0, 0\}$ since we start in the intracellular state. The emission probabilities vary by state:

$$e_I(a) = \frac{1}{20} \quad \forall a \in \Sigma$$

$$e_T(a) = \begin{cases} \frac{15}{100}, & a \in \{L, I, V, F, W\} \\ \frac{1}{100}, & \text{otherwise} \end{cases}$$

$$e_E(a) = \begin{cases} \frac{43}{300}, & a \in \{R, K, D, E, Q, N\} \\ \frac{1}{100}, & \text{otherwise} \end{cases}$$

The transition probabilities are given by the edges in the diagram, which can be represented in matrix form as follows:

$$T = \begin{bmatrix} 0.6 & 0.4 & 0 \\ 0.2 & 0.6 & 0.2 \\ 0 & 0.4 & 0.6 \end{bmatrix}$$

Problem 2:

- (a) The theoretical residue mass is: 781.349 Da.
The theoretical peptide mass is: 799.365 Da.
The ion mass for charge $Z = 2$ is: 400.682 Da.

(b)